Determinants and Effects of Community Participation in Road Paving with Labor Based Technology: A Case Study at Nong Ko Village, Thailand

LBT（労働集約型）道路舗装事業への参加要因と参加の影響：タイ・ノンコー村の事例分析

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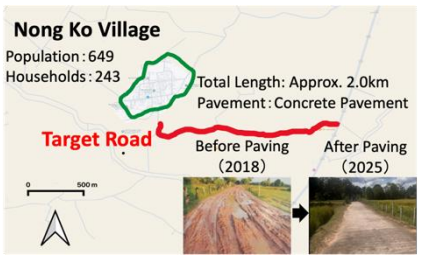

1 Background

Challenges of community participation in LBT projects

- Heavily rely on external support for completion
- Inadequate maintenance due to low participation

LBT road paving project in Nong Ko Village, Thailand

- Successful completion without government support
- Donation and physical labor by villagers





Research question

What are the determinants of successful LBT participation in Nong Ko Village?

2 Goals

This study aims to:
(1) Identify the determinants of community participation;
(2) Assess the effects of participation on villagers' subsequent awareness of community participation;
(3) Discuss policy implications

3 Methods

Mixed-methods: SEM and Semi-structured interviews

Model Analysis

- Structural Equation Model (SEM)
- Questionnaire survey data from 355 villagers, collected in 2019 (Takeda, 2023)

Field Surveys

- Field visit at Nong Ko Village in 2026
- Interviews with 17 villagers
- Identification of participation mechanism

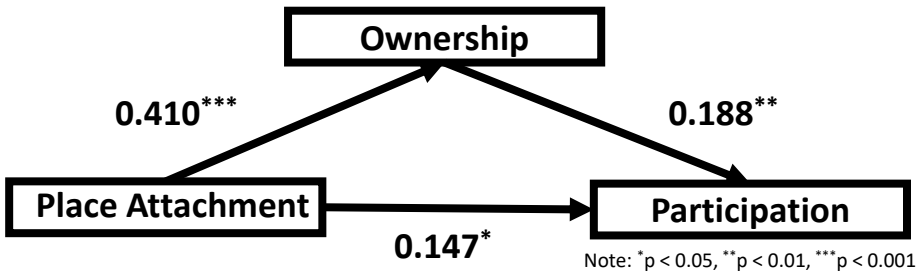
Discussion on policy implications

4 Results

Model Estimation Results

- Place attachment has a positive effect on community participation in the LBT road paving project
- Ownership partially mediates the relationship

Estimated participation model



Note: *p < 0.05, **p < 0.01, ***p < 0.001

Cultural values in the village

Local interviews unveiled:

- **Trust in the village head**

"The village head shares all details and never deceives the villagers. This is exactly why I feel a sincere desire to participate. " (70s, female)



- **The spirit of *Tambun***

"… I thought of it as a way to help develop the village and make travel more convenient, while as a form of making merit" (80s, male)



Effects of LBT participation

- **Future participation willingness**

"In the future, these collective efforts can be applied to other fields to foster village development" (50s, male)



5 Conclusions

- Place attachment and ownership affect community participation rooted in cultural values.
- LBT serves as an effective strategy for the sustainable development of rural infrastructure.

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